



Analysis of Bridges & Wormholes Paper

Part 4 – Study Summary

First Delivered: 02 Jun 2026

Last Updated: 02 Jun 2026

Michael M Lee

Review of Study Work Products



Presentations:

- Part 1 Consider Control (mint.wets.tn.7)
- Part 2 Consider SW Architecture Requirements
plus Control Shortcomings and Recommendations (mint.wets.tn.8)
- Part 3 Consider Data Handling,
plus Shortcomings and Recommendations (mint.wets.tn.9)
- Part 4 Study Summary (mint.wets.tn.10)

Annotated B&W Paper (pdf):

- SW Architecture and Data Handling references highlighted (mint.wets.tn.11)

Available on Modelint GitHub:

<https://github.com/modelint/wets-case-study/wiki/Document-Register>



Assessment and Shortcomings Summary

- The Bridges & Wormholes approach works in the context presented

Shortcomings

- The approach does not support:
 - a blocking, synchronous CRP service without return data, or
 - maintaining identifier mapping half tables when the instance population is dynamic, or
 - a place for logic to address semantic conversions of parameters.
- The approach is limited in that it requires CRPs to be synchronous or asynchronous which restricts how a wormhole/client can use them.
- The approach makes identifier mapping half tables unnecessarily complex by constructing them at the domain level.



Recommendations for Control Shortcomings

1. Support a blocking, synchronous CRP service without return data
2. Support a CRP providing both synch and asynch behavior

To Do That:

- A. Have separate request wormholes for synch/asynch behavior
 - Makes HOME behavior explicit in action language
- B. Introduce a Return Mechanism supertype for Return Coordinates and Transfer Vectors subtypes*
 - The CRP can handle both without knowing the difference.
- C. Have the CRP invoke a Return Wormhole after completing its service and provide the Return Mechanism and any return data items.
 - The CRP does not need to know whether its service was provided synchronously or asynchronously. The software architecture handles that.

[*] Andrew suggests handling return information at the ME, not the CRP, level. Great idea!

Recommendations for Data Handling Shortcomings



1. Clarify that there are distinct data item types in a Wormhole/CRP signature
 - Separate guidance for handling each type can be provided
2. Address handling identifier mapping when the instance populations are dynamic.
 - Not sure how to do this in general, although solutions for specific application behaviors have been proposed by both Mike & Andrew.
3. Create Identifier Mapping Half Tables at the CRP (not Domain) level
 - Simplifies the effort by decomposing the problem
4. Provide a place for logic to address semantic conversions of parameters?
 - Does not seem possible in current B&W approach

Successes (+) and Failures (-) for Improving Shortcomings



Shortcomings

- The approach does not support:
 - a blocking, synchronous CRP service without return data, or +
 - maintaining identifier mapping half tables when the instance population is dynamic, or + / -
 - a place for logic to address semantic conversions of parameters. -
- The approach is limited in that it requires a CRP to be synchronous or asynchronous which restricts how a wormhole/client can use them. +
- The approach makes identifier mapping half tables unnecessarily complex by constructing them at the domain level. +

My \$.02 on Bridges & Wormholes Paper



Given the limited automation tooling available to explore and validate approaches to bridging and the fact that this was their first (and only) complete pass at addressing bridging, I think the B&W paper made a significant step forward in

- + identifying the requirements for supporting explicit bridging and proposing how to address them. That said, it fell short in the clarify of presentation and the completeness of solution -
- relative to their other method expositions. I would guess that it was the result of a best effort in a constrained timeframe. +/ -



Given the limited automation tooling available to explore and validate approaches, this was their first (and only) bridging, I think the B&W effort in identifying the bridging and proposing a solution all short in the clarify of problem relative to their solution - guess that it was the result of a best effort in a constrained timeframe. +/ -

